

CHENG-WEN FU

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EDUCATION

Carnegie Mellon University, Pittsburgh, PA 05/2027
Master of Science in Computational Design

National Cheng Kung University, Tainan, Taiwan 06/2023
Bachelor of Architectural Design

WORK EXPERIENCE

Modeler and Computational Designer, *Illoca, US* 04/2024 - 08/2025

- Assisted in building architectural models and developing parametric façade systems using Revit and Grasshopper, contributing to the creation of AI training datasets for design prediction and optimization.

Computational Designer, *J. J. Pan and Partners, Architects & Planners, TW* 05/2024 - 04/2025

- Applying computational technologies to support design development, such as using Flow Designer and Ladybug to inform sustainable design strategies, developed custom plugins to meet project needs, and contributed to the writing and production of the firm's annual publication.

Research Assistant, *Civil Engineering Dept. National Taiwan University, TW* 12/2023 - 07/2025

- Developed web-based BIM visualization and data-driven tools, focusing on interactive 3D rendering, clear data visualization, and streamlined user workflows. Collaborated on system architecture and data-flow design to ensure smooth integration across models, interfaces, and backend services.

Architectural Intern, *Field Office Architects, TW* 07/2021 - 01/2022

- Assisted in conceptual and physical development of an urban design proposal, producing detailed drawings, study models, and client-presentation materials.

PROJECT EXPERIENCE

Interactive Web Developer, *Web-Based BIM Visualization Platforms* 08/2025

- Designed a visualization interface powered by Autodesk APS Simple Viewer, enabling real-time 3D exploration of formwork inventory and component movement patterns, supporting logistics and inventory management.

Interactive Web Developer, *SDGs Campus Information Integration Platform* 02/2025

- Built an integrated BIM with IoT digital twin platform using Autodesk APS APIs, enabling 3D model versioning, sensor registration, and real-time data mapping across multiple buildings.

Tool Developer, *Building Embodied Carbon Calculator* 01/2025

- Designed and implemented a custom Grasshopper plugin for early-stage carbon-emission analysis in collaboration with green-building consultants, integrating data visualization and comparative metrics to help designers evaluate the carbon performance of design schemes.

PUBLICATION

First Author, *International Conference on Civil and Building Engineering Informatics* 01/2025

- Created a streamlined workflow and interface that simplifies digital twin deployment and data maintenance, reducing implementation cost and enhancing building energy management toward net-zero performance.
- [Fu., Wu., Chang., and Hsieh. "Streamlining Sensor Registration and Updating Process in a BIM-based Digital Twin." Proceedings of the International Conference on Civil and Building Engineering Informatics \(ICBEI\), 2025.](#)

SKILLS

Coding: Python, JavaScript, HTML/CSS, React, Node.js, PostgreSQL

Modeling, Visualization & Immersive Tool: Rhino, Grasshopper, Revit, Enscape, Adobe PS, Adobe AI, Adobe ID, Figma

Environmental & System Simulation: Flow Designer, Climate Studio